

CS & IT ENGINEERING



Computer Network

IPv4 Address

Lecture No. - 03

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Recap of Previous Lecture



Topic

Broadcast Address

Topic

Network Size



Topics to be Covered



Topic

Network Mask

Topic

Classful IPv4 Address

ABOUT ME



Hello, I'm **Abhishek**

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Example 6 :-

[MCQ]



#Q. Consider a class-less IPv4 network, the broadcast address (special IPv4 address) of this network is "175 . 175 . 175 . 255" and size of network identifier field is 22 bits, which one of the following can be a network address (special IPv4 address) of this network ?

- A 175 . 175 . 0 . 0
- B 175 . 175 . 160 . 0
- ☒ C 175 . 175 . 172 . 0
- D 175 . 175 . 175 . 0

Ans: C

Network Directed Broadcast Address = 175.175.175.255

10101111 10101111 10101111 11111111
 22 bit NetID 10 bit HostID

Network Address =

10101111 10101111 10101100 00000000
 22 bit NetID 10 bit HostID
175.175.172.0

[NAT]

[GATE-2015][2 Mark]



#Q. In the network "200 . 10 . 11 . 144 / 27", the fourth octet (in decimal) of the last IP address of the network which can be assigned to a host is _____.

144 $\rightarrow$ $\underbrace{100}_{\text{Network ID}} \underbrace{0000}_{\text{5 bit Host ID}}$

Ans $\rightarrow$ $\underbrace{100}_{\text{Network ID}} \underbrace{11110}_{\text{5 bit Host ID}}$

Ans = 158

IPv4 Address = 200.10.11.144 / 27

= 200.10.11.10010000
 27 bit NetID 5 bit HostID

Last IP Address that can be assigned to a host =

200.10.11.10011110
 27 bit NetID 5 bit HostID
 Last IP Address = 200.10.11.158
 [that can be assigned to a host]



Topic : Network Mask



→ Network Mask (Netmask) [32-bits]

NetID field = All One Bits

HostID field = All Zero Bits

$a \rightarrow \text{bit}$

$$a \text{ AND } 1 = a$$

$$a \text{ AND } 0 = 0$$

Network Id [1 1 1 1 1 1 1]

Host Id [0 0 0 0 0 0 0]

$x \text{ bit}$

$y \text{ bits}$

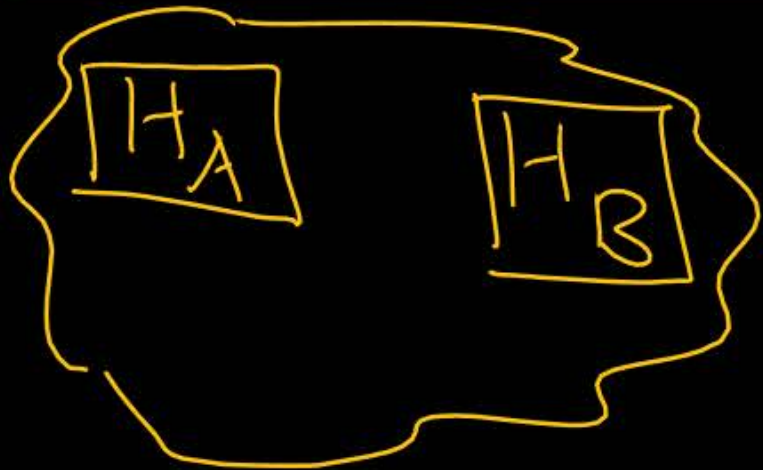


Topic : Network Mask

→ Binary string used to produce “Network Address” of an IP address

Network Address ← IP Address [bit-wise AND] Netmask

→ All hosts belongs to same network must have same “Network Address” * *



$$\begin{array}{l} IP_A \\ \text{(bitwise AND)} \\ \text{NetMask} \\ \hline \text{Net Add}_1 \end{array}$$

$$\begin{array}{l} IP_B \\ \text{(bitwise AND)} \\ \text{NetMask} \\ \hline \text{Net Add}_1 \end{array}$$

$$\begin{array}{l} IP_C \\ \text{(bitwise AND)} \\ \text{NetMask} \\ \hline \text{Net Add}_2 \end{array}$$





Topic : Network Mask



IP Address

:

Network Id

Host Id

bitwise AND

[bit - wise AND]

Network Mask

:

1 1 1 1 1 1 1

0 0 0 0 0 0 0 0

Network Address

:

Network Id

0 0 0 0 0 0 0 0

Example 10 :-

[MCQ]



#Q. Consider a class-less IPv4 network where the size of network identifier field is 22 bits, which one of the following is network mask of this network ?

- A 255 . 255 . 0 . 0
- B 255 . 255 . 248 . 0
- ✓ C 255 . 255 . 252 . 0
- D 255 . 255 . 255 . 0

Ans: C



Network Mask = 11111111 11111111 11111100 00000000

22 bit NetID 10 bit HostID

Netmask = 255 . 255 . 252 . 0

Example 11 :-

[NAT]



#Q. Consider a class-less IPv4 network, the network mask of this network is "255 . 255 . 248 . 0", how many maximum number of hosts can be in this network ?

Network Mask = 255.255.248.0

= 11111111 11111111 11110000 00000000

21-bit NetID 11-bit HostID

Number of bits in Host ID field (y) = 11

Maximum number of hosts can be in a network = $[2^y - 2]$ hosts per network

Ans = 2046

= $(2^{11} - 2)$ hosts
 = 2046 hosts

Example 12 :-

[MSQ]



#Q. Consider two host belongs to a class-less IPv4 network, the IP addresses of the hosts are "178 . 158 . 138 . 118" and "178 . 168 . 158 . 148", which is/are can be a network mask of this network ?

- ☒ A 255 . 0 . 0 . 0
- ☒ B 255 . 192 . 0 . 0
- ☐ C 255 . 240 . 0 . 0
- ☐ D 255 . 255 . 0 . 0

Ans: A & B



$$\begin{aligned} 178, 158, 138, 118 &= 178 \cdot 10011110 \cdot 138 \cdot 118 \\ 178, 168, 158, 148 &= 178 \cdot 10101000 \cdot 158 \cdot 148 \end{aligned}$$

A

178.158.138.118
255. 0. 0. 0

[bitwise AND]

178.168.158.148
255. 0. 0. 0

178. 0. 0. 0

178. 0. 0. 0

B

178.158.138.118
255.192. 0. 0

178.168.158.148
255.192. 0. 0

178.128. 0. 0

178.128. 0. 0

168 → 10101000
192 → 11000000
128 ← 10000000

178 → 10110010
255 → 11111111

178 ← 10110010

158 → 10011110
0 → 00000000

0 ← 00000000

158 → 10011110
192 → 11000000

128 ← 10000000

C

178.158.138.118

255.240. 0. 0

178.144. 0. 0

178.168.158.148

255.240. 0. 0

178.168. 0. 0

D

178.158.138.118

255.255. 0. 0

178.158. 0. 0

178.168.158.148

255.255. 0. 0

178.168. 0. 0

158 → 10011110
240 → 11110000
144 ← 10010000

168 → 10101000
240 → 11110000
160 ← 10100000

[MCQ]

H.W.

(IIT-G)

[GATE-2010][2 Mark]



#Q. Suppose computers A and B have IP addresses "10 . 105 . 1 . 113" and "10 . 105 . 1 . 91" respectively and they both use the same netmask N. Which of the values of N given below should not be used if A and B should belong to the same network?

A

255 . 255 . 255 . 0

B

255 . 255 . 255 . 128

C

255 . 255 . 255 . 192

D

255 . 255 . 255 . 224

[MCQ]

Netmask (H.W.)

[IT-M]

[GATE-2003][2 Mark]



#Q. The subnet mask for a particular network is "255 . 255 . 31 . 0". Which of the following pairs of IP addresses could belong to this network ?

- A 172 . 57 . 88 . 62 and 172 . 56 . 87 . 233
- B 10 . 35 . 28 . 2 and 10 . 35 . 29 . 4
- C 191 . 203 . 31 . 87 and 191 . 234 . 31 . 88
- D 128 . 8 . 129 . 43 and 128 . 8 . 161 . 55



2 mins Summary



Topic

Network Mask

Topic

~~Classful IPv4 Address~~



THANK - YOU

